

## Activity IV - Fundamental of Cryptography

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### Overviews

In this activity, we will learn the basics of encryption. There are 3 exercises in this activity. Each exercise is designed to let you learn the concepts of cryptography.

We will need:

- Imagemagick โหลดเอา
- OpenSSL เปิด wsl แล้วใช้ openssl เอา
- One of your favorite programming languages.

You are welcome to do this exercise with any programming language. If you have no preference, use python.

### Exercises

1. (Cryptanalysis) Though encryption is primarily designed to preserve confidentiality and integrity of data, the mechanism itself is vulnerable to brute force (statistical analysis). In other words, the more we see the encrypted data, the easier we can hack it. In this exercise, you are asked to crack the following cipher text. Please provide the decrypted result and explain your strategy in decrypting this text.

#### **Cipher text**

PRCSOFQX FP QDR AFOPQ CZSPR LA JFPALOQSKR. QDFP FP ZK LIU  
BROJZK MOLTROE.

- a. Count the frequency of letters. List the top three most frequent characters.
- b. Knowing that this is English, what are commonly used three-letter words and two-letter words. Does the knowledge give you a hint on cracking the given text?
- c. Cracking the given text. Measure the time that you have taken to crack this message.
- d. Create a simple python program for cracking the Caesar cipher text using brute force attack. Explain the design and demonstrate your software.  
(You may use an English dictionary for validating results.)

2. (Cryptanalysis on Symmetric Encryption) Vigenère is a complex version of the Caesar cipher. It is a polyalphabetic substitution. วิธี hack vigenere

- a. Please review **Kasiski examination**<sup>1</sup> Explain how it can be used to attack Vigenère.

hint: ถ้าจะเขียน program เพื่อ hack ใอนี้ hack อย่างไร ข้อ 1 กับข้อ 2 ขอแค่ มี processing power ดีๆ hack ได้เสมอ

3. (Mode in Block Cipher) Block Cipher is designed to have more randomness in a block. However, an individual block still utilizes the same key. Thus, it is recommended to use a cipher mode with an initial vector, chaining or feedback between blocks. This exercise will show you the weakness of **Electronic Code Book** mode which does not include any initial vector, chaining or feedback.

- a. Find a bitmap image that is larger than 2000x2000 pixels. Note that you may resize any image. To simplify the pattern, we will change it to bitmap (1-bit per pixel) using the portable bitmap format (pbm). In this example, we will use imagemagick for the conversion. ดัด header ออกป้องกันดูไม่ได้ เพราะ header โดนencrypt

```
$ magick convert image.jpg -resize 2000x2000 org.pbm
```

- b. The NetPBM<sup>2</sup> format is a naive image format. The first two lines contain a header (format and size in pixel). Depending on the format, the pixels can be represented in either binary and ascii. For our exercise, we prefer binary. However, we first have to take out the header to prevent the encryption from encoding the header. To do so, use your text editor (eg. vi, notepad) to take out the first two lines.

```
$ cp org.pbm org.x
```

```
$ vi org.x
```

```
P4
```

```
2000 2000
```

```
KR)B@HD♦♦@H♦
```

รูปต้องใหญ่ระดับนี้ และ detail ต้องใหญ่ๆ หยิบมาๆ

- c. Encrypt the file with OpenSSL<sup>3</sup> with any block cipher algorithm in ECB mode (no padding and no salt).

```
$ openssl enc -aes-256-ecb -in org.x -nosalt \
-out enc.x
```

- d. Pad the header back and see the result.

```
$ cp enc.x enc.pbm
```

```
$ vi enc.pbm
```

```
P4
```

```
2000 2000
```

```
KR)B@HD♦♦@
```

- e. You may try it with other modes with IV, chaining, or feedback and compare the result.

- f. What does the result suggest about the mode of operation in block cipher? Please provide your analysis.

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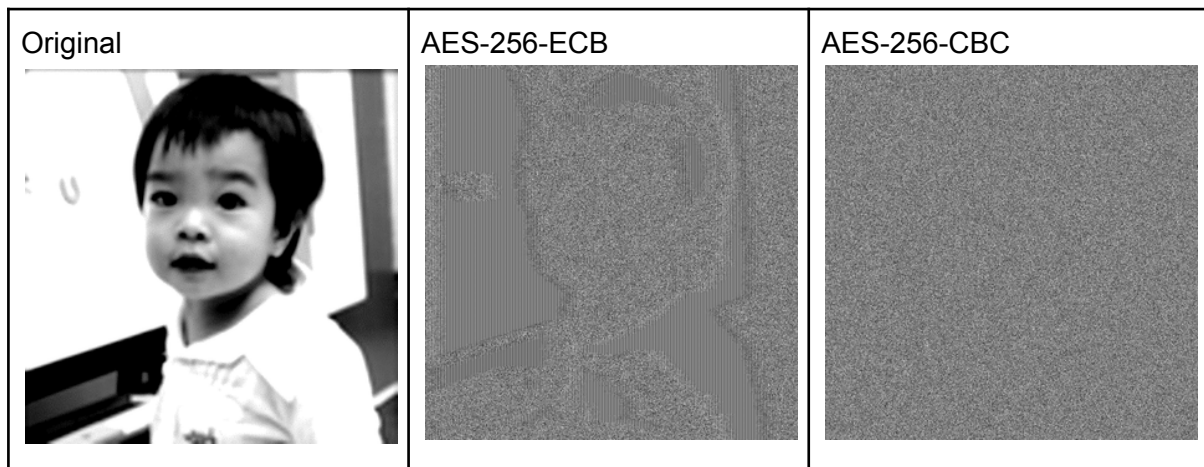
<sup>1</sup> See wikipedia for more details. [https://en.wikipedia.org/wiki/Kasiski\\_examination](https://en.wikipedia.org/wiki/Kasiski_examination)

<sup>2</sup> See wikipedia for more details. <https://en.wikipedia.org/wiki/Netpbm>

<sup>3</sup> For details on command-line arguments, see <https://wiki.openssl.org/index.php/Enc>

If you got it all right, the result should be like this.

หารูปมาใส่กรุป แล้วเข้ารหัสสองแบบนี้ แล้วดูว่ามี pattern มั้ย



4. (Encryption Protocol - Digital Signature)

- Measure the performance of a hash function (sha1), RC4, Blowfish and DSA. Outline your experimental design. (Please use OpenSSL for your measurement)
- Comparing performance and security provided by each method.
- Explain the mechanism underlying Digital Signature. How does it combine the strength and weakness of each encryption scheme?

หาข้อความ text file ใหญ่ๆ 1 mb 2mb แล้ว encrypt ด้วย แบบต่างๆและ compare เวลา

Hint: (OpenSSL command line)

# List algorithms

```
$ openssl list cipher-algorithms
```

# To encrypt

```
$ openssl enc -ciphername [options] -e -in filename -out filename \  
-K key -iv IV -nopad -nosalt
```